Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program . This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Enabled Multi-Legged Robots for Inspection and Intervention in Challenging Environments			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Robot Control and Intelligence Technologies	*Robotics/ Automation Machinery*	Robot Vision and Production Automation Technologies
Concept of Proposed Technology	The proposed technology aims to develop an AI-driven, multi-legged robot platform designed to serve as a force multiplier for human operators in "Dangerous, Dull, and Dirty" (3D) environments. While the primary pilot application focuses on **in-home pest control**, the underlying technology—comprising learning-enabled locomotion and multimodal perception—is built for general-purpose utility in **agriculture, defense, and industrial inspection**. The platform's novel, centipede-like design allows it to confidently AND RELIABLY (CITE SCIENCE PAPER) navigate cluttered and complex terrains that defeat wheeled, tracked, and few-legged			

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	robots. Traversable environments include tall grass, rocky debris, mud, pinestraw, ice, and stairs. In the pest control context, the system is not intended as a standalone consumer appliance. Instead, it functions as a **technician-aid tool**, entering hazardous and confined spaces such as **attics and crawlspaces** (characterized by insulation, joists, tarps, and debris) to serve as the operator's remote "eyes, ears, and arms". AI is leveraged to: 1. **Augment Locomotion:** Ensuring stable traversal over unpredictable terrain. 2. **Enhance Detection:** Providing technicians with automated tools to identify pest presence, building damage, or structural anomalies through fused multimodal signals. Autonomous locomotion is also being developed for the specialty agriculture market, where robots will act as "vineyard managers" that automatically patrol fields, using AI-assisted cameras and sensors to monitor crop health while also deterring agricultural pests.
Rationale for Joint R&D Project	The proposed project requires close collaboration between a research institute, a faculty-founded startup (Ground Control Robotics Inc.), and a Korean industrial partner to translate AI-enabled multi-legged mobility into a professional-grade tool for sectors involving the "3Ds": **Dangerous, Dull, and Dirty**. While broadly applicable to agriculture and defense, this project focuses on a practical in-home pest control system designed to assist human operators. Unlike consumer "cleaning robots," this platform is intended to work alongside a **pest control technician** to access tough-to-reach and hazardous spaces—such as **attics and crawlspaces**—serving as the operator's remote eyes, arms, and ears in environments that are unsafe or difficult for humans to navigate. A joint R&D framework is essential to co-develop and integrate AI algorithms with ruggedized robotic hardware capable of traversing **challenging domestic constraints**. These include unstructured environments like fiberglass insulation, narrow joists, debris-filled crawlspaces, and variable flooring where traditional wheeled robots fail. This collaboration will ensure the system provides the technician with AI-driven tools to better identify pest presence and building damage, facilitating a transition from lab-based robophysics to a reliable, field-deployable professional service robot. **Rationale for collaboration:** • **Research institute + Ground Control Robotics Inc.** will lead the core algorithm and research efforts, including learning-based locomotion, tactile sensing, and closed-loop autonomy. This will focus on "challenging indoor navigation" scenarios, specifically the high-complexity traversal required for attics and crawlspaces where traditional sensing is often occluded.

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	• **Korean SME partner** Will drive product-oriented engineering and **manufacturing at scale**. Their role includes system packaging, embedded integration, and the design of professional intervention modules—such as trap/treatment payloads—optimized for reliability and field evaluation in representative homes. They will also assist with developing AI tools for pest and damage detection to augment the robotic platform. • **Tight hardware–software co-design** is necessary to satisfy practical deployment constraints, including cost, safety, noise, cleaning/maintenance, durability, and usability for non-expert users. **Expected benefits of joint R&D:** • **Accelerated Professional Deployment:** Facilitates the transition from research prototypes to ruggedized, deployable systems specifically designed for professional technicians to use in hazardous in-home environments like attics and crawlspaces. • **De-risking of 3D Operations:** Reduced technical risk via iterative co-development and validation in realistic mock-ups, ensuring the robot can safely navigate "Dangerous, Dull, and Dirty" spaces that pose health or safety risks to human operators. • **Scalable Foundation for Professional Service Robotics:** Establishes a commercialization pathway for service robots designed for professional home sanitation and pest management. Rather than a consumer-facing appliance, the platform serves as a specialized tool for pest control companies, enabling technicians to safely and effectively inspect and intervene in hazardous or tough-to-access spaces like attics and crawlspaces. This model leverages AI not only to ensure robust locomotion in unstructured environments but also to empower operators with advanced diagnostic tools for identifying pest presence and structural damage.
Key Words	1. Robotics/robophysics 2. Terraydynamics 3. Bioinspired locomotion 4. Autonomous locomotion 5.Autonomous Inspection and Monitoring

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (5)	End of the Project	Level (9)
R&D Objectives	The primary R&D objective is to develop a robust, AI-enabled multi-legged robot platform that serves as a professional-grade tool for technicians, providing reliable traversal and diagnostic capabilities in hazardous, "3D" (Dangerous, Dull, and Dirty) environments such as attics and crawlspaces. The specific R&D objectives are: • To develop robust locomotion and tactile-first navigation algorithms enabling reliable traversal in complex, unstructured spaces, including insulation-filled attics and debris-heavy crawlspaces. • To develop AI-based pest event detection and localization tools that assist professional operators in identifying likely infestation hotspots and structural damage under partial observability. • To develop AI-enabled intervention behaviors and a technician-in-the-loop coordination framework for targeted responses, such as deploying traps or verifying hotspots in hazardous zones.			
R&D Target Specifications	**Hazardous Environment Traversal (≥95%):** Achieve ≥95% traversal success across representative "3D" obstacles—including loose fiberglass insulation, uneven ceiling joists, and wet crawlspace surfaces—while maintaining stable operation and preventing entrapment. **Technician-Assisted Inspection Coverage (≥90%):** Achieve ≥90% coverage of specified high-risk inspection zones (e.g., HVAC duct perimeters, deep attic corners, and structural sills) within a standard 30-minute service window, providing the operator with a comprehensive digital map of the area. **Augmented Diagnostic Accuracy (≥85%):** Achieve ≥85% precision in identifying and localizing pest-related events (e.g., entry points, droppings, or nesting) and building damage (e.g., wood rot or wire chewing) by fusing tactile, acoustic, and visual data for the technician's review. **Operational Durability & Serviceability (≥100 hours):** Achieve a Mean Time Between Failure (MTBF) of ≥100 operational hours in high-dust and high-moisture environments, featuring a modular hardware design that allows a technician to perform field maintenance or sensor cleaning in under 10 minutes.			
R&D Plan	**Year 1: Simulation-Based Development and Core AI Technology Implementation** • Develop an integrated simulation environment for multi-legged indoor traversal and inspection tasks • Implement learning-based locomotion and tactile-first navigation policies for confined spaces			

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	• Implement multi-modal event detection and hotspot inference (tactile + optional minimal vision / acoustic / vibration) • Validate core algorithms through simulation benchmarks and stress tests **Year 2: Laboratory-Scale Integration and System Prototyping** • Integrate locomotion, sensing, and detection modules on a multi-legged robot platform in a lab “mock home” environment • Develop intervention modules (e.g., trap/treatment payload prototype, docking/charging, maintenance workflow) • Customize and optimize AI algorithms for real hardware constraints (sensor noise, drift, wear, contamination) • Conduct repeated lab experiments to evaluate robustness, safety, and task success rates **Year 3: Real-Home Deployment and Technology Advancement** • Deploy the integrated system in realistic home environments (pilot trials) • Perform large-scale experiments to verify traversal, inspection coverage, pest detection/localization, and intervention performance • Refine system reliability, user experience, and operational safety based on experimental results • Assess commercialization potential and prepare the technology for industrial transfer (manufacturing and certification readiness)

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute, in close coordination with the faculty-founded startup **Ground Control Robotics Inc.**, will be responsible for developing the core AI algorithms and system-level autonomy for the multi-legged robot. Together, they will lead research on perception and control methodologies required for reliable operation in **hazardous and confined 3D environments** (e.g., attics and crawlspaces) and will support the translation of these methods into integrated robotic prototypes. **Specific roles and responsibilities include:** • **Learning-Based Locomotion & Navigation:** Development of tactile-first navigation and inspection algorithms tailored for high-complexity, unstructured environments such as loose insulation, narrow joists, and debris-filled crawlspaces. • **Augmented Diagnostic AI:** Development of pest event detection and structural damage localization tools that provide professional technicians with enhanced situational awareness under partial observability. • **Autonomy Stack Design:** Design of the overall autonomy framework, including the integration of sensing-control-inspection loops with a technician-in-the-loop interface. • **Simulation-Based Validation:** Large-scale simulation benchmarking to de-risk hardware deployment and accelerate algorithm iteration for "Dangerous, Dull, and Dirty" scenarios. • **Technical Integration Support:** Providing guidance to the Korean SME partner on experimental specifications, joint debugging, and real-home pilot evaluations.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for development, integration, and validation of **robot hardware**, **intervention modules**, and **product-level system implementation** required for in-home deployment. Specific roles and responsibilities include: • **Hardware Ruggedization & Manufacturing:** Design and **large-scale manufacturing** of multi-legged robot hardware, including protective enclosures for high-dust/moisture environments and serviceability features for field technicians. • **Specialized Intervention Modules:** Development of professional-grade intervention mechanisms, such as precision trap/treatment payloads and safe interaction tools. • **Advanced Perception Augmentation:** Development of AI tools for pest and damage detection that augment the base robot's capabilities. • **System Integration:** Integration of AI algorithms with embedded software, focusing on reliability engineering and power management for extended field operations. • **Field Evaluation:** Execution of laboratory-scale and real-home pilot trials, focusing on performance data collection, cost reduction, and commercialization readiness for the professional service market.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to deliver a professional-grade, multi-legged AI-enabled robotic platform capable of reliable traversal and diagnostic intervention in hazardous domestic environments. By transitioning from a consumer appliance concept to a **technician-aid tool**, the project will advance the state of the art in tactile-first indoor robotics and learning-based locomotion specifically for the **"3Ds": Dangerous, Dull, and Dirty**. **Expected Outcomes and Impact** • **Technician Safety and Efficiency:** The primary impact is the significant reduction of human exposure to hazardous conditions in attics and crawlspaces. By serving as a remote extension of the technician, the system improves inspection thoroughness and operational safety. • **Next-Generation B2B Service Model:** This project establishes a scalable Business-to-Business (B2B) foundation for the commercialization of service robots in the pest management and home inspection industries. • **Industrial Competitiveness for Korean SMEs:** The project provides the Korean SME partner with proprietary technologies in **manufacturing at scale** for multi-legged systems and AI-driven diagnostic tools. These outcomes will support the partner's growth in the global AI-driven service robotics market, extending to applications in **agriculture and defense**. • **Technological Leadership:** The project will yield measurable improvements in traversal success (≥95%) and pest/damage detection accuracy (≥85%) in unstructured environments, setting a new benchmark for "tactile-first" robotics in the industry.

6. References

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(3) He, Juntao, Baxi Chong, Jianfeng Lin, Zhaochen Xu, Hosain Bagheri, Esteban Flores, and Daniel I. Goldman. "Probabilistic approach to feedback control enhances multi-legged locomotion on rugged landscapes." IEEE Transactions on Robotics (2025).

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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